



Mechanical Engineering (ME)

University of Michigan College of Engineering

ME 495 – Laboratory II

Lab 2: Flexible Torsional System

(Scenario)

Wolverine System Dynamics
Consultants for Automotive, Aero, and Space Technology
Advanced Industry Research Park
Ann Arbor, MI 48109



To: Powertrain Dynamics Group
From: Edgar Meyhofer and Pramod Reddy,
Directors, Wolverine System Dynamics
Date: Sept 4th, 2026
Subject: Drive Shaft Vibrations in the Gulo 3000

During the last regular planning meeting with Mesa Hybrid Vehicles Inc., a major vibration problem was identified in its new prototype hybrid electric model, the *Gulo 3000*. The design team believes that this vibration is associated with the drive shaft connecting the DC motor to the front-wheels of the car. The vibration manifests itself transiently when the DC motor is turned on/off and was found to be sufficiently large to be uncomfortable to the driver and passengers. The general conclusion from our meeting was that the magnitude of the vibration was unacceptable and needs to be reduced by at least 30%.

From discussions with team members of Mesa's *Gulo* design group, it appears that the "flexibility" of the drive-axle, which was assumed to be rigid in their design work, is the cause of the undesirable vibrations. Fortunately, we have available in-house, from previous work, an appropriate physical model of the motor and drive-axle system that has been instrumented and is accessible for your use in the Systems Lab. Your group is assigned the task of first developing a motor/drive-axle model to better understand the system and to subsequently remedy the vibration problem.

Specifically, you need to

- (1) validate that the vibration is indeed due to shaft flexibility,
- (2) characterize the system and develop a mathematical model, and
- (3) perform simulations using the validated model to identify possible design changes to solve the vibration problem.

Mesa would like to review several possible solutions to the vibration problem and select, based on a broad range of criteria beyond the actual drive shaft design, the best design for the final powertrain. Therefore, you are charged with identifying the four most effective solutions to achieve a 30% reduction in the vibrations.

This project has an extremely short timeline and must have your highest priority. You will be presenting the results of your experimental work and analysis in a poster session in the week of September 21. Because of the significance of this vibration problem, the poster session will be attended by Mesa engineers and executives with diverse technical backgrounds. Please ensure that your poster uses clear visual elements to provide a brief, broadly comprehensible introduction and conclusions alongside the central technical data. In addition, a short, three-page handout summarizing the key findings from your work is due to us in the same week during your poster presentation.

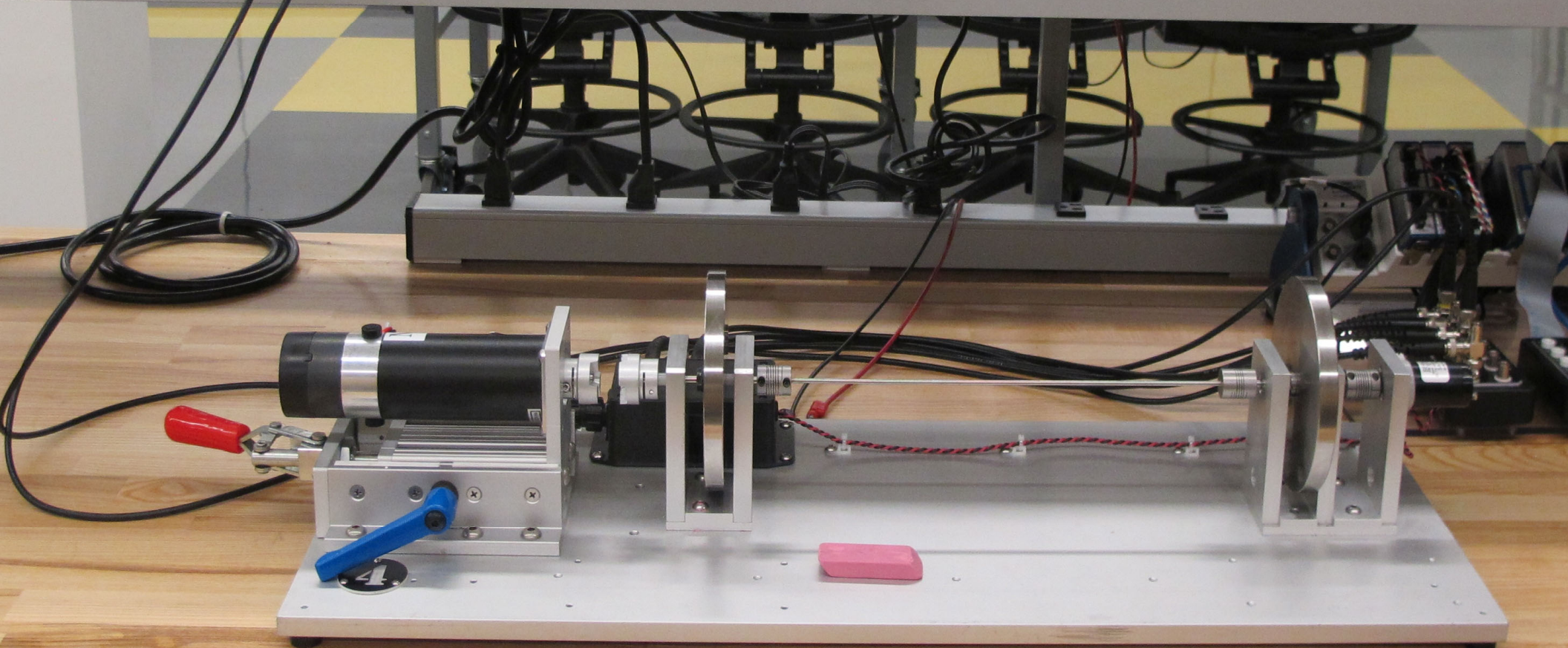
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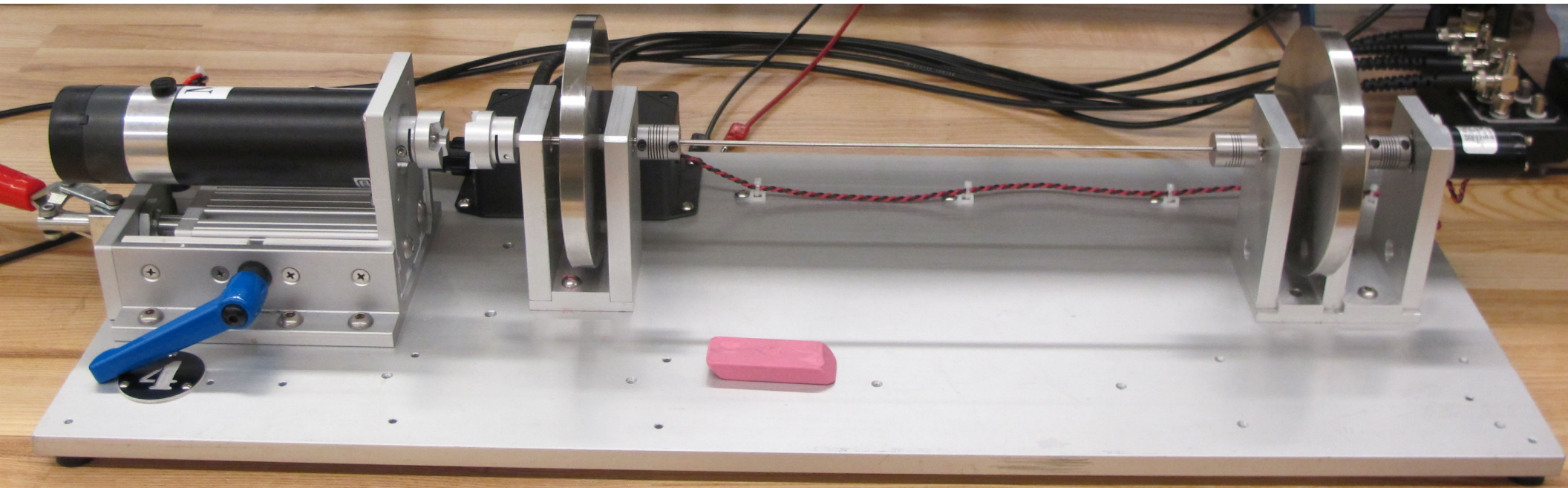
Problem and Main Objectives of Lab 2

- **Dynamical model of automotive drive train**
- **Goal, reduce resonance (peak magnitude ratio)**
- **First, develop a model to better understand the problem**
- **Verify that model agrees with test data**
 - if not, find out why and obtain an improved model**
- **Finally, use the model to develop a new design that will address the resonance problem. What is the best solution?**

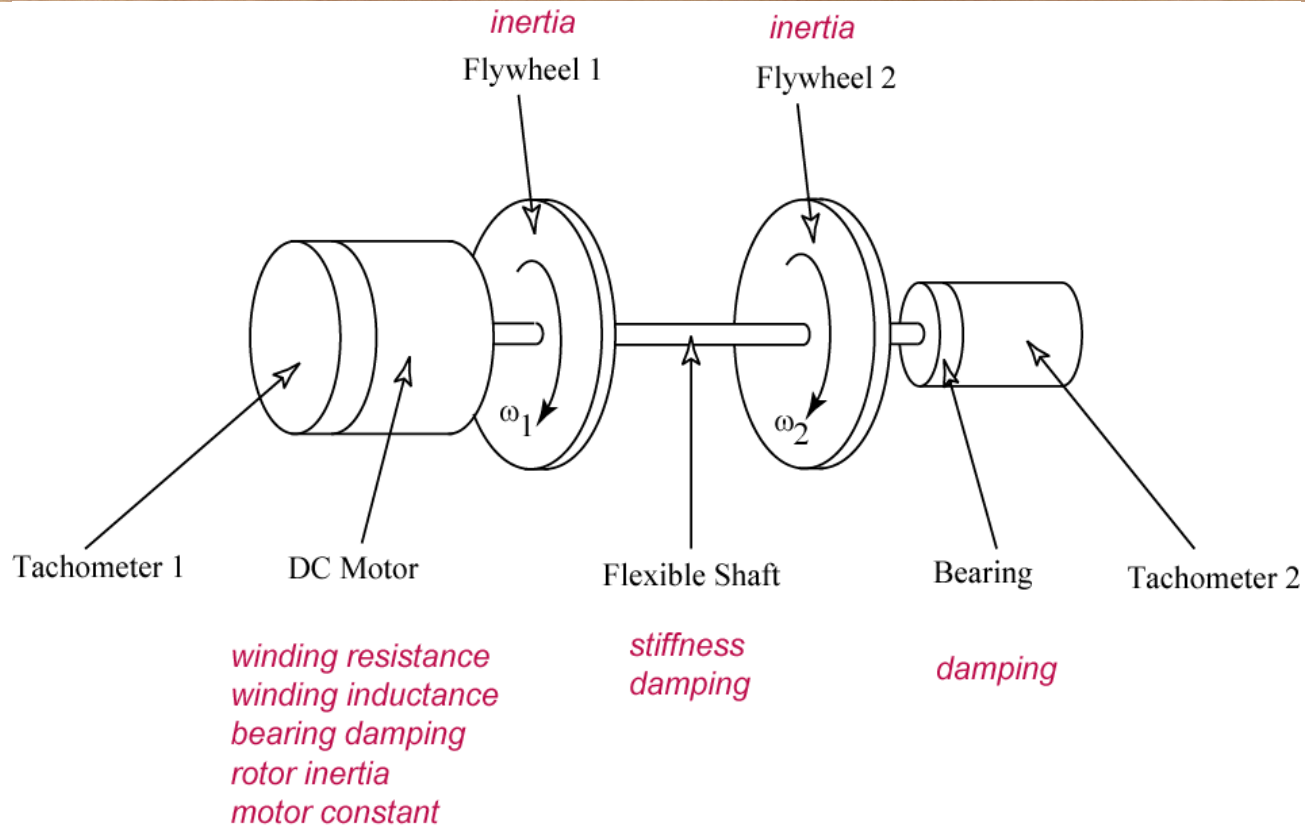
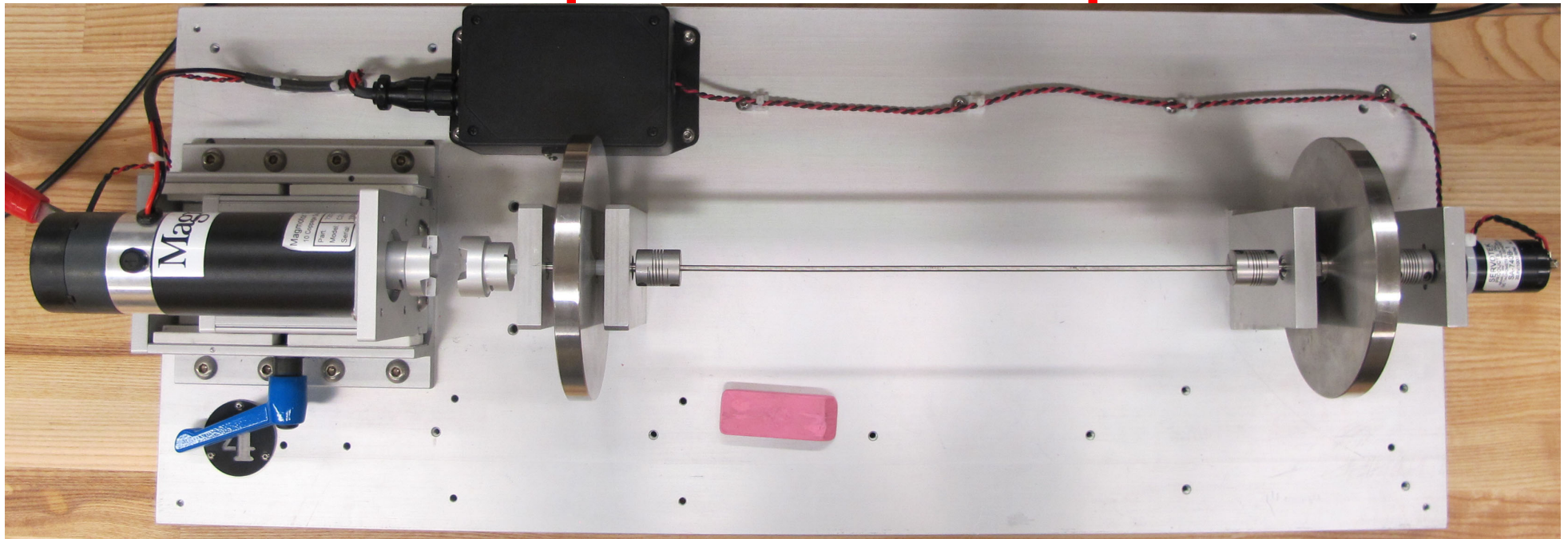
Goals for the First Week of Measurements (Lab)

- **Team Formation (new teams)**
- **Collecting frequency response data and other required data and parameters to build a mathematical model**
- **Implement a first model**
- **Identify possible weaknesses of model, and develop a plan for its improvement in the following week**

Experimental Set-Up

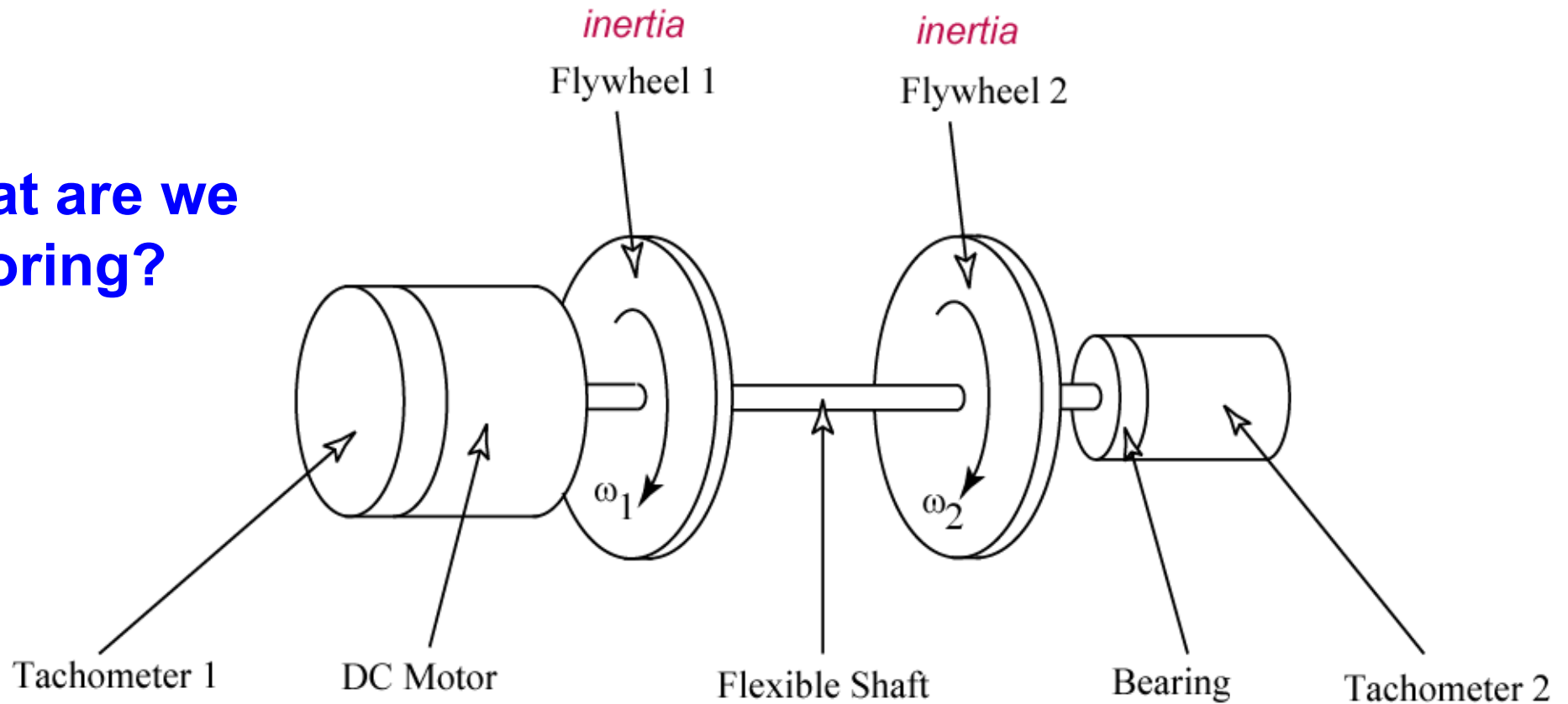


Experimental Set-Up



Flexible Torsional System Schematic

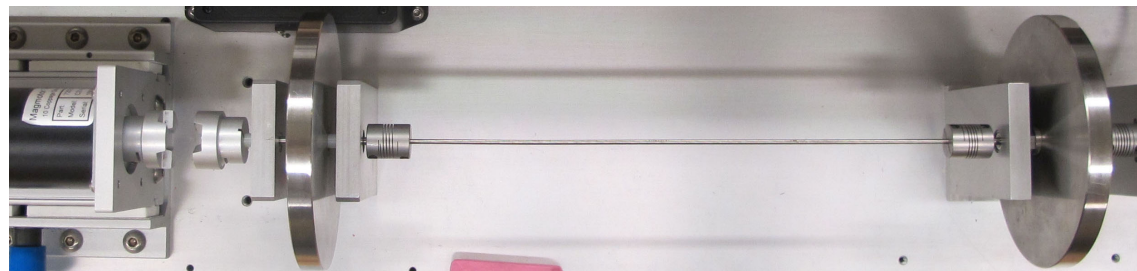
What are we ignoring?



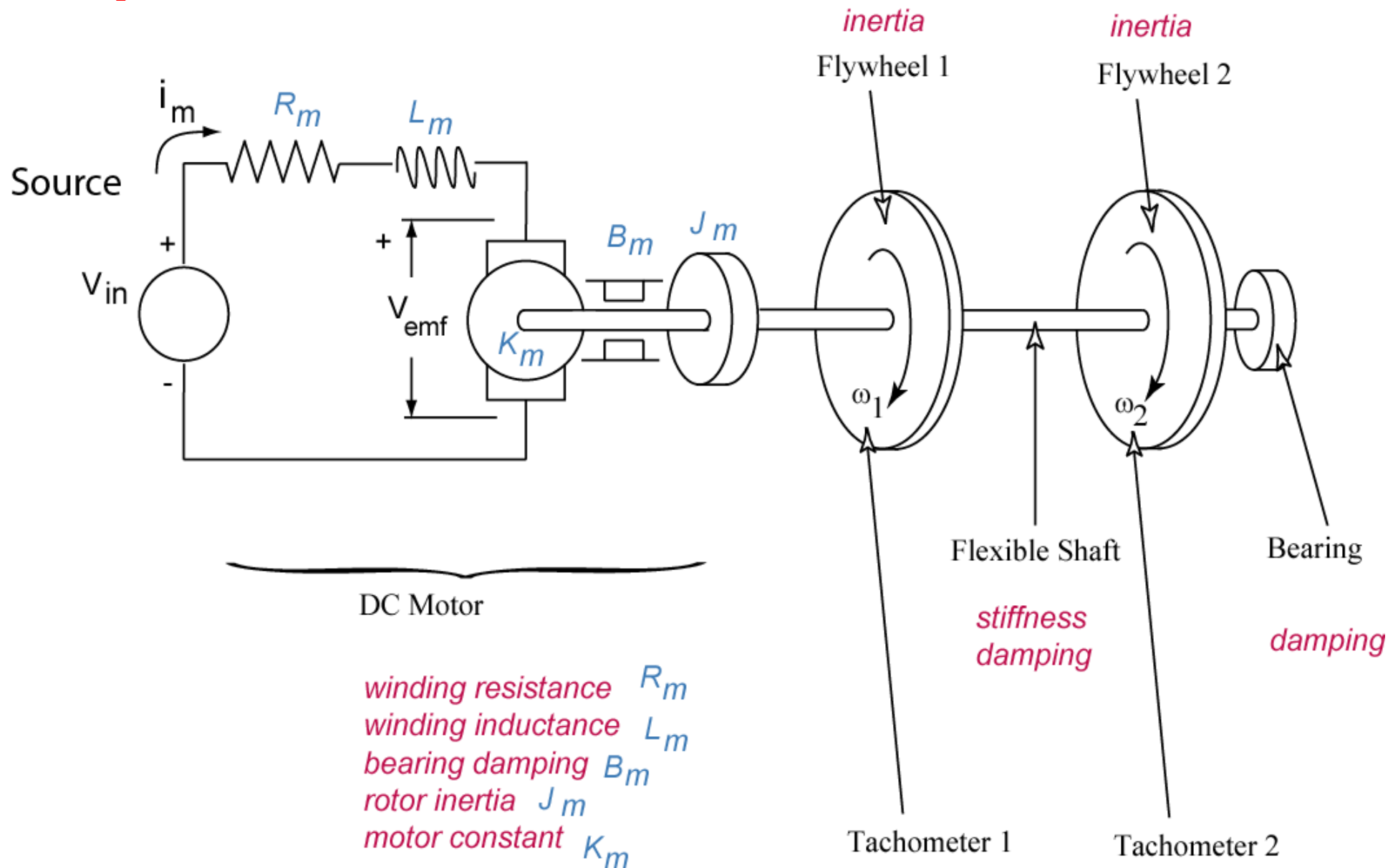
winding resistance
winding inductance
bearing damping
rotor inertia
motor constant

stiffness
damping

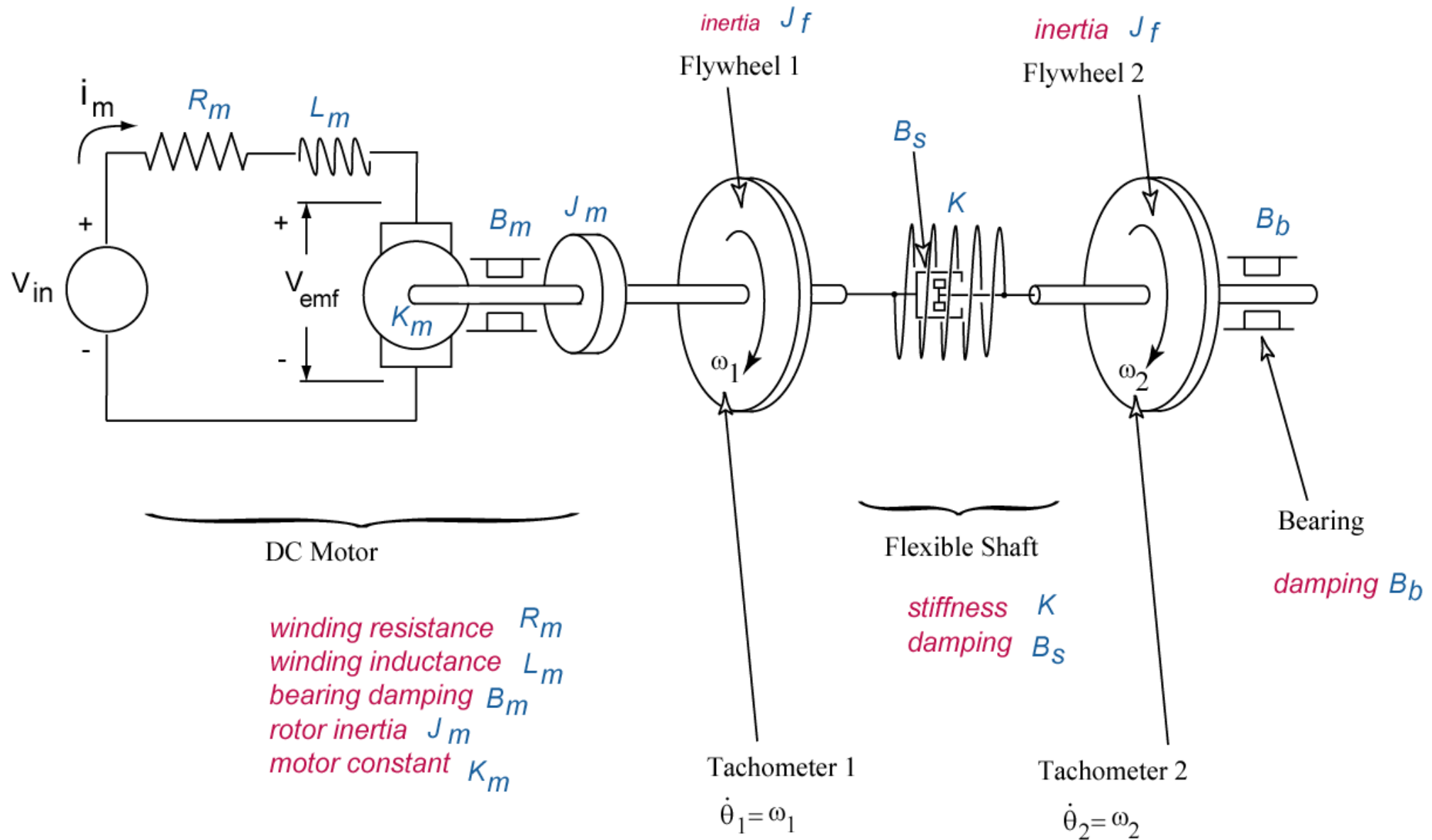
damping



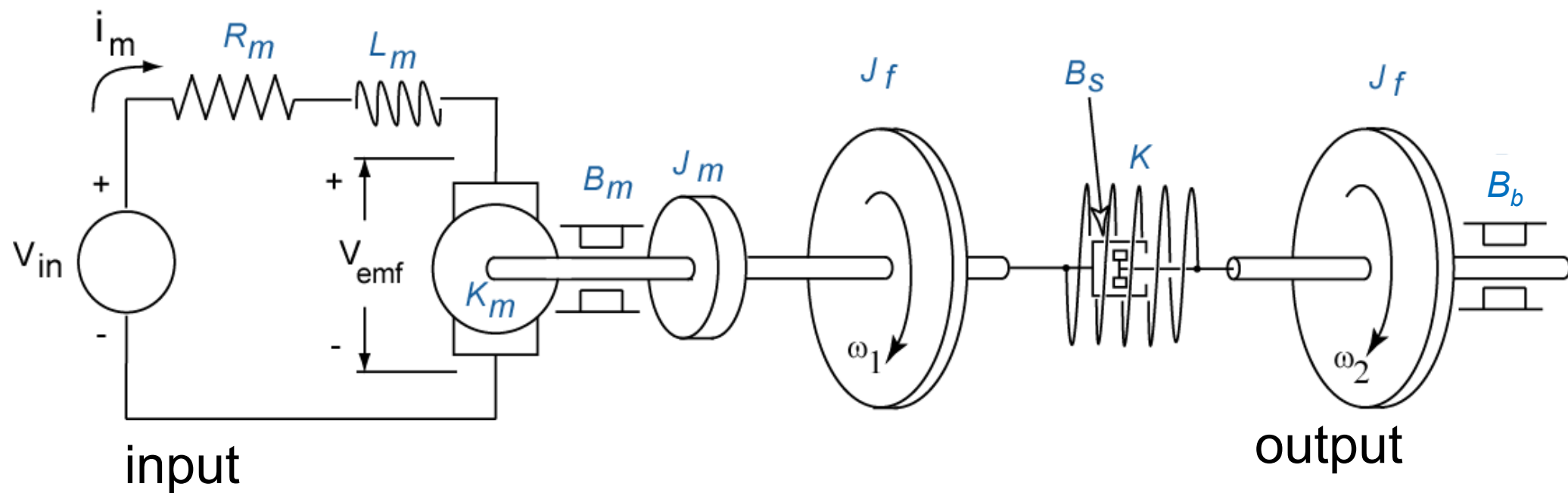
Schematic with DC Motor Representation



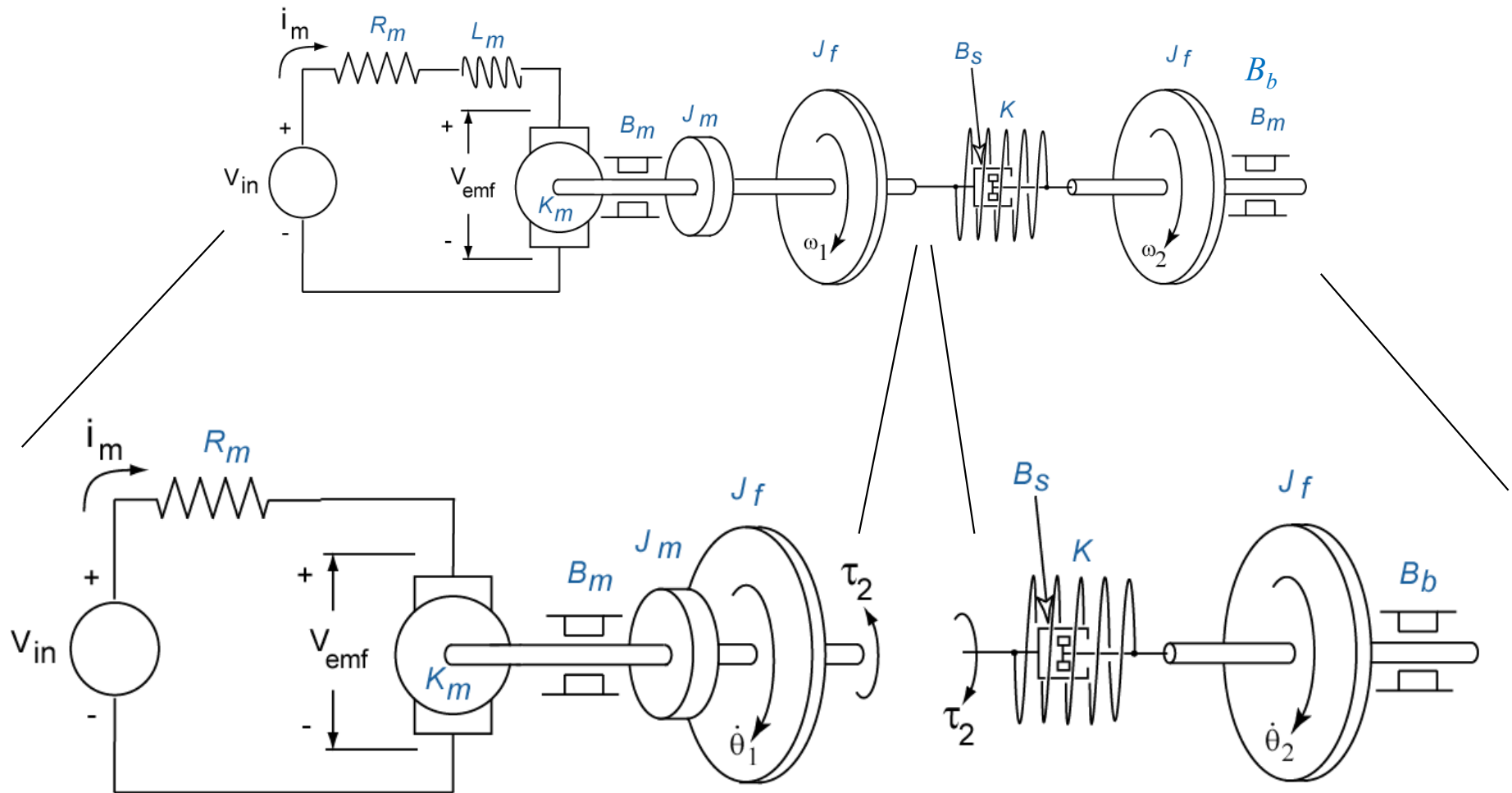
Full System Schematic



Goal: Develop a Transfer Function: $\frac{\dot{\theta}_2(s)}{V_{in}(s)}$

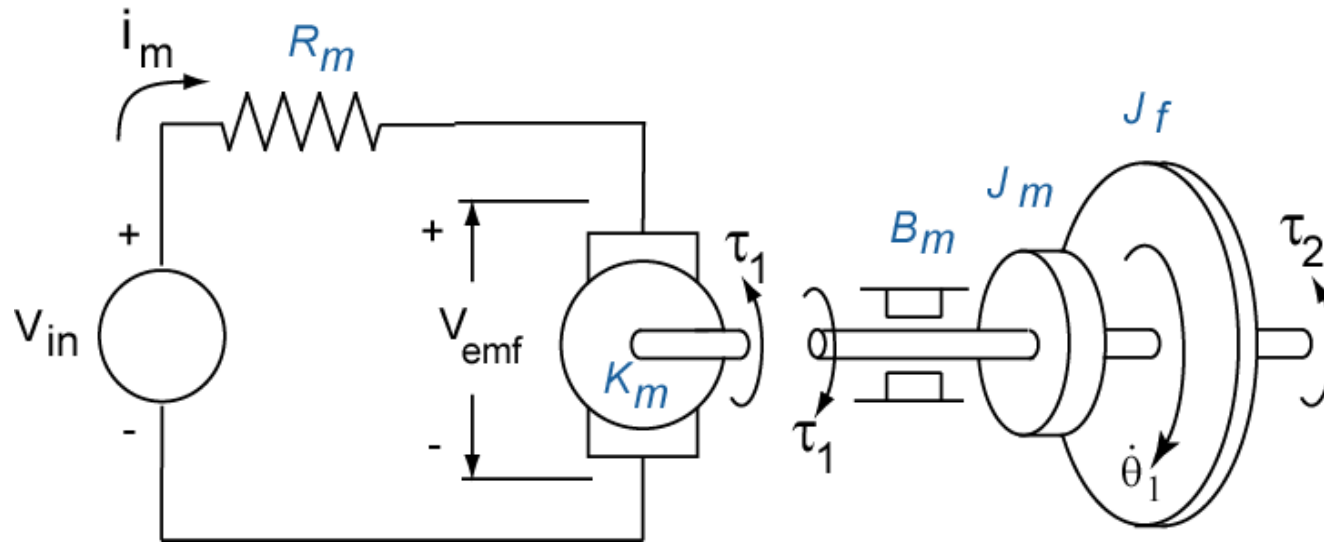


To develop Equations of Motion: (Modeling) Simplify the System



Modeling (cont.)

Governing Equations



$$V_{in} = R_m i_m + V_{emf}$$

$$V_{emf} = K_m \dot{\theta}_1$$

Back EMF

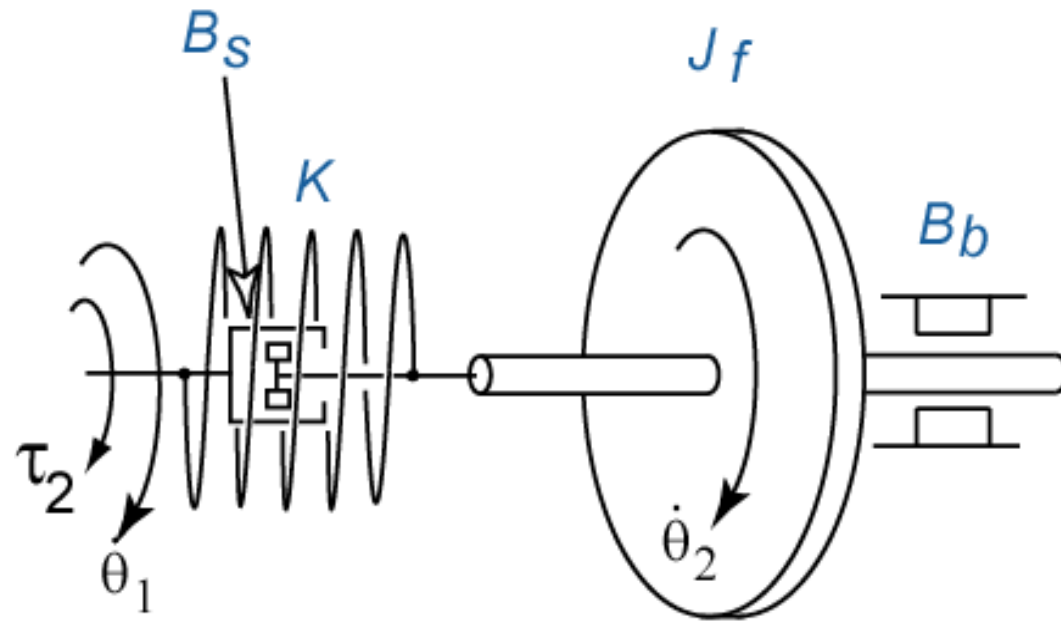
$$\tau_1 = K_m i_m$$

Motor Torque

$$\tau_1 = \tau_2 + B_m \dot{\theta}_1 + (J_m + J_f) \ddot{\theta}_1$$

Modeling (cont.)

Shaft + Flywheel #2



$$\tau_2 = B_s (\dot{\theta}_1 - \dot{\theta}_2) + K (\theta_1 - \theta_2)$$

$$\tau_2 = J_f \ddot{\theta}_2 + B_b \dot{\theta}_2$$

Transfer Function

Take Laplace transform of equations

Eliminate all intermediate variables.

$$\tau_2 = B_s (\dot{\theta}_1 - \dot{\theta}_2) + K (\theta_1 - \theta_2)$$

$$\tau_2 = J_f \ddot{\theta}_2 + B_b \dot{\theta}_2$$

$$\tau_2(s) = B_s [\dot{\theta}_1(s) - \dot{\theta}_2(s)] + \frac{K}{s} [\dot{\theta}_1(s) - \dot{\theta}_2(s)]$$

$$\tau_2(s) = [J_f s + B_b] \dot{\theta}_2(s)$$

$$V_{in}(s) = \frac{R_m \tau_2(s)}{K_m} + \left[\frac{R_m J_c s}{K_m} + K_m + \frac{R_m}{K_m} B_m \right] \dot{\theta}_1(s)$$

$$J_c = J_m + J_f$$

Transfer Function (cont.)

$$H(s) = \frac{\dot{\theta}_2(s)}{V_{in}(s)}$$
$$= \frac{K_m(B_s s + K)}{R_m J_c J_f s^3 + [R_m J_c B_{tr} + J_f(K_m^2 + B_{tl} R_m)] s^2 + \left\{ R_m [K(J_c + J_f) + B_b B_s] + (K_m^2 + R_m B_m) B_{tr} \right\} s + K [K_m^2 + (B_b + B_m) R_m]}$$

where

$$B_{tr} = B_s + B_b$$

$$B_{tl} = B_s + B_m$$

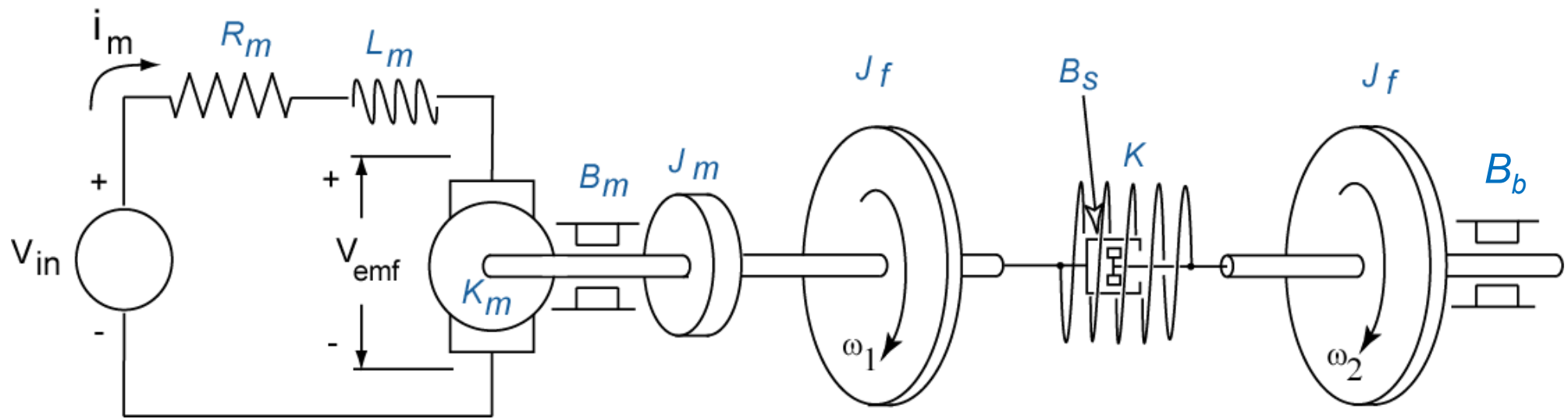
$$J_f = \frac{\rho_f \pi D_f^4 L_f}{32}$$

$$K = \frac{G_s J_s}{L_s} = \frac{\pi G_s D_s^4}{32 L_s}$$

Obtain and Verify All Model Parameters

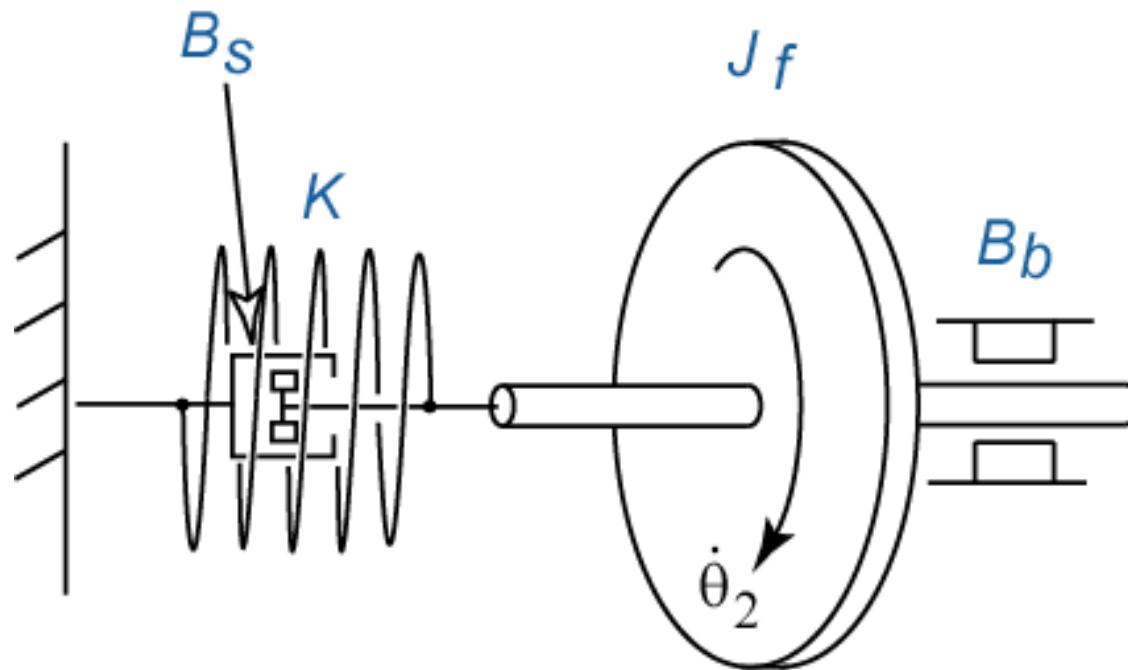
DC Motor		
Motor constant	$K_m = 0.097$ $K_m = 0.06$	MagMotor [N·m/Amp] or [V/(rad/sec)] Aerotech [N·m/Amp] or [V/(rad/sec)]
Winding resistance	$R_m = 1.6$ $R_m = 0.9$	MagMotor [ohms] Aerotech Motor
Winding inductance	$L_m = 0.002$	[Henry]
Rotor inertia	$J_m = 4.23 \times 10^{-5}$	MagMotor [kg·m ²]
2 Flywheels		
Diameter	$D_f = 0.137$	[m]
Length (width)	$L_f = 0.0127$	[m]
Density	$\rho_f = 7755$	[kg/m ³]
Shaft		
Diameter	$D_s = 3.18 \times 10^{-3}$	[m]
Length (width)	$L_s = 0.305$	[m]
Shear modulus	$G_s = 7.31 \times 10^{10}$	[N/m ²]
Tachometer #1		
Scale factor	7.0 (MagMotor) 3.0 (Aerotech)	[V/KRPM]
Tachometer #2		
Scale factor	20.8	[V/KRPM]

Which model parameters are not specified?



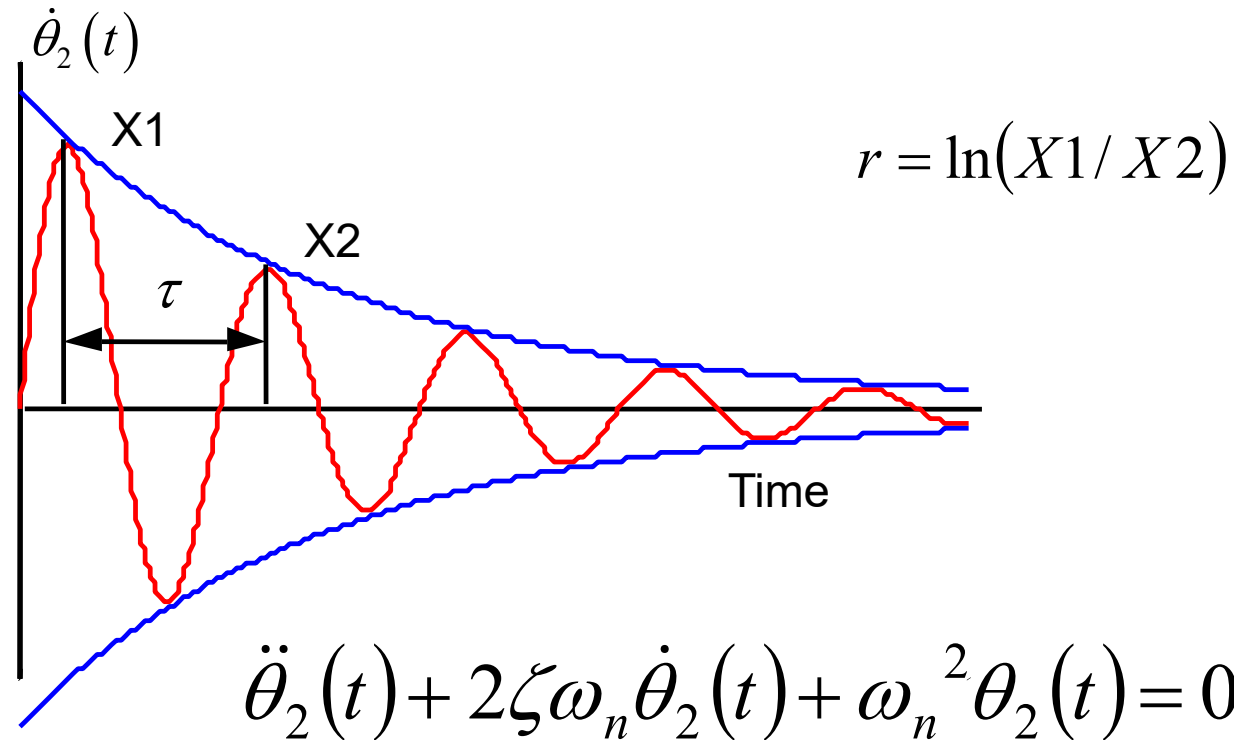
Obtain Values of Unspecified Damping Parameters

Example for B_{tr} : How do we obtain the damping coefficients?



$$J_f \ddot{\theta}_2(t) + B_{tr} \dot{\theta}_2(t) + K \theta_2(t) = 0$$

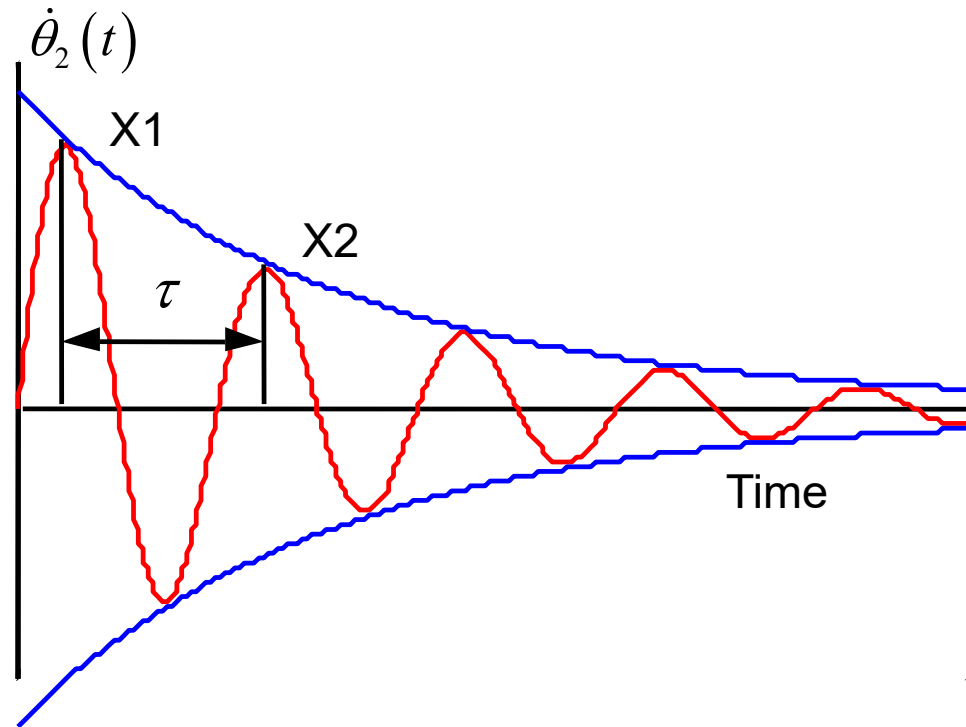
Free Response of 2nd Order Systems



$$\theta_2(t) = \theta_2(0)e^{-\zeta\omega_n t} \left[\frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_d t + \cos \omega_d t \right] \quad \omega_d = \omega_n \sqrt{1-\zeta^2}$$

$$\dot{\theta}_2(t) = -\frac{\theta_2(0)}{1-\zeta^2} e^{-\zeta\omega_n t} \sin \omega_d t$$

Free Response of 2nd Order Systems



$$r = \ln(X1 / X2) \quad \text{Logarithmic decrement}$$

$$\zeta = \frac{r}{\omega_n \tau}$$

**Damping ratio,
dimensionless**

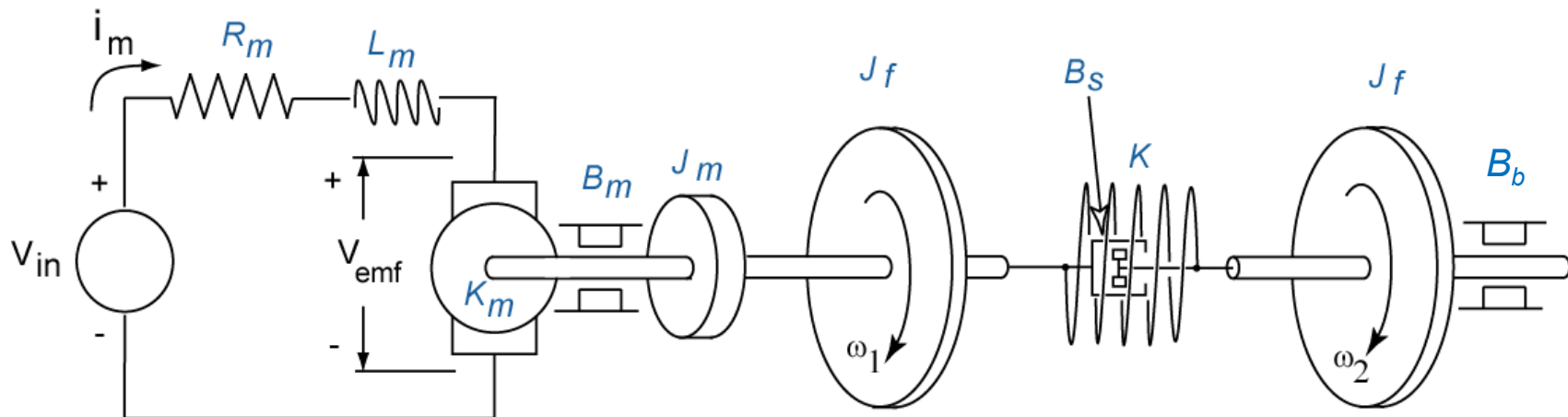
$$\zeta = \frac{\text{damping coefficient}}{\text{critical damping}}$$

Lab 2 Week 1

Procedure to Estimate Damping Coefficients

- Lock flywheel #1
- Give flywheel #2 a slight twist
- Record free response on oscilloscope
- Compute the ratio 'r'. This calculation should be done by using the experimental response of several cycles to reduce the effect of noise and other uncertainties. Once 'r' is known, calculate ζ . (Consult a Vibration Textbook as needed)
- From ζ calculate the actual damping coefficient

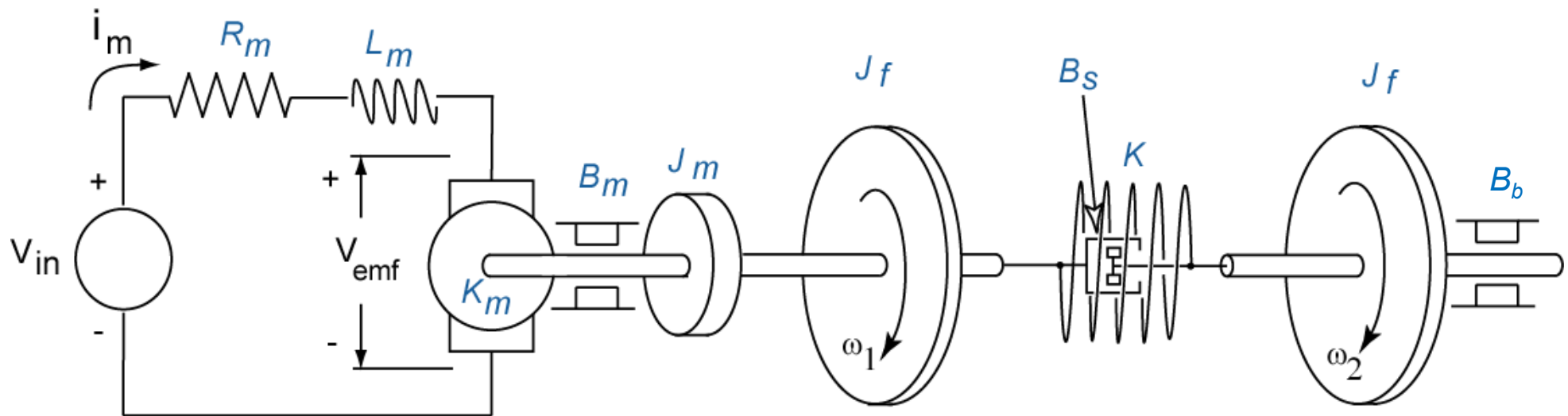
$$r = \ln(X1 / X2)$$



Lab 2 Week 1 Procedure

Estimate Damping Coefficients

- Design a second test procedure to obtain the damping coefficient B_{tl}
- Potential improvements: Design an additional test procedure to determine all three coefficients. If you choose to skip this step, you can assume $B_m = 0$ and attribute half of B_{tr} to B_s and the other half to B_b . If you choose other ways to divide up, please justify.

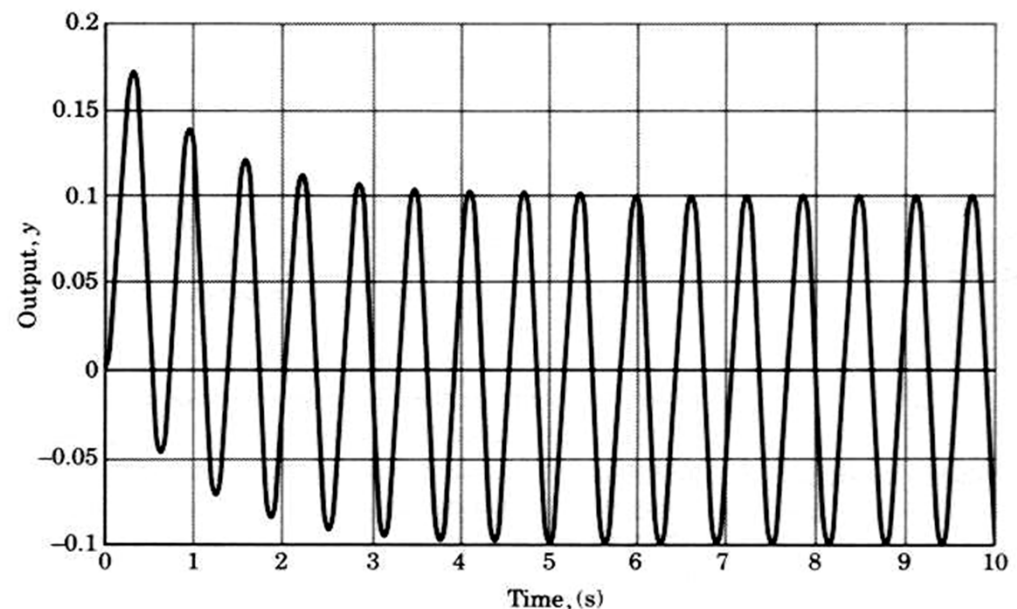


Summary: Lab 2 Week 1 Procedure

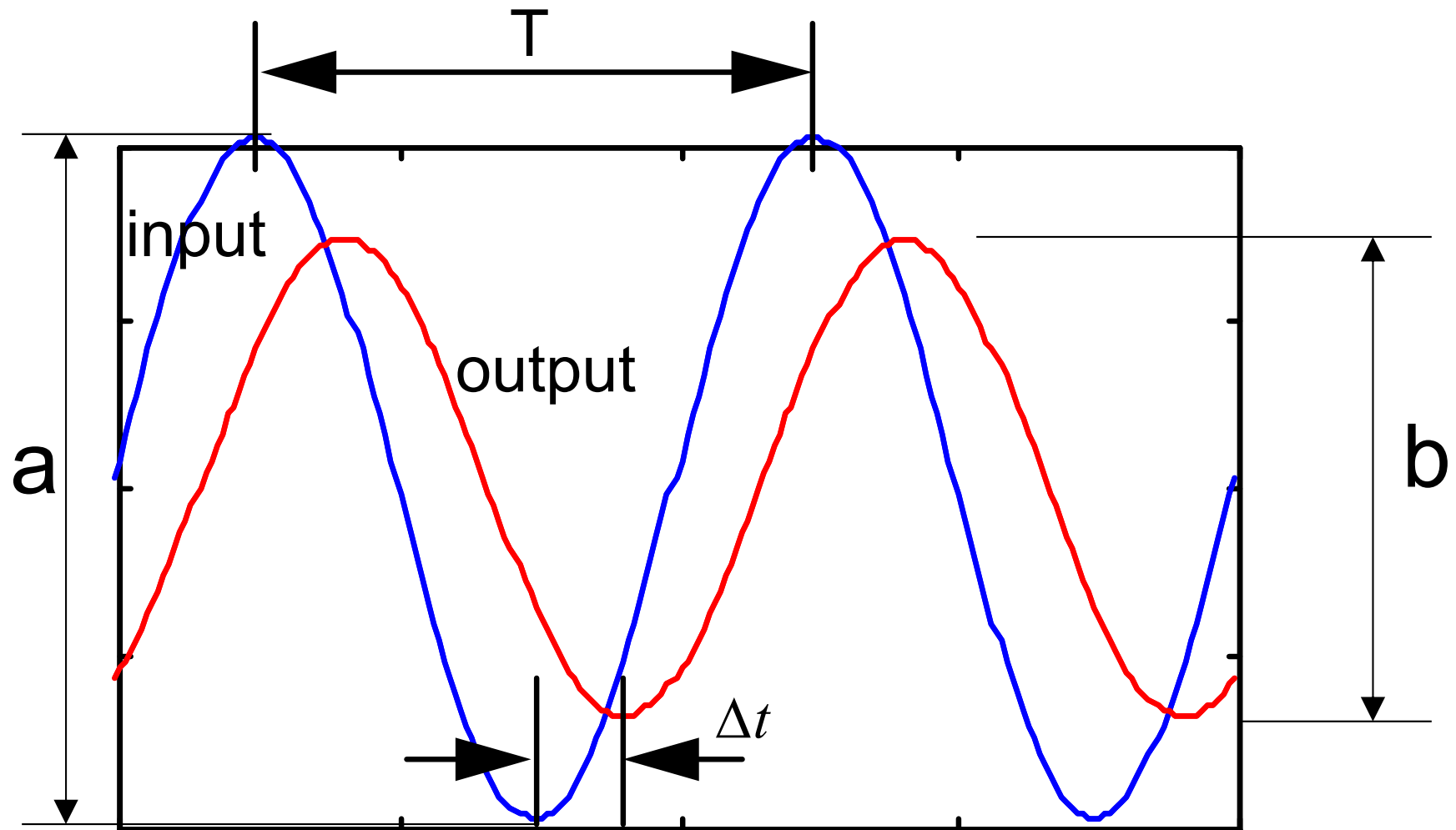
- Experimentally Determine the Frequency Response of the System
- Identify Damping Coefficients
 - as detailed above
- Implement a first mathematical model describing the system behavior and evaluate if your model is appropriate

Reviewing Frequency Response

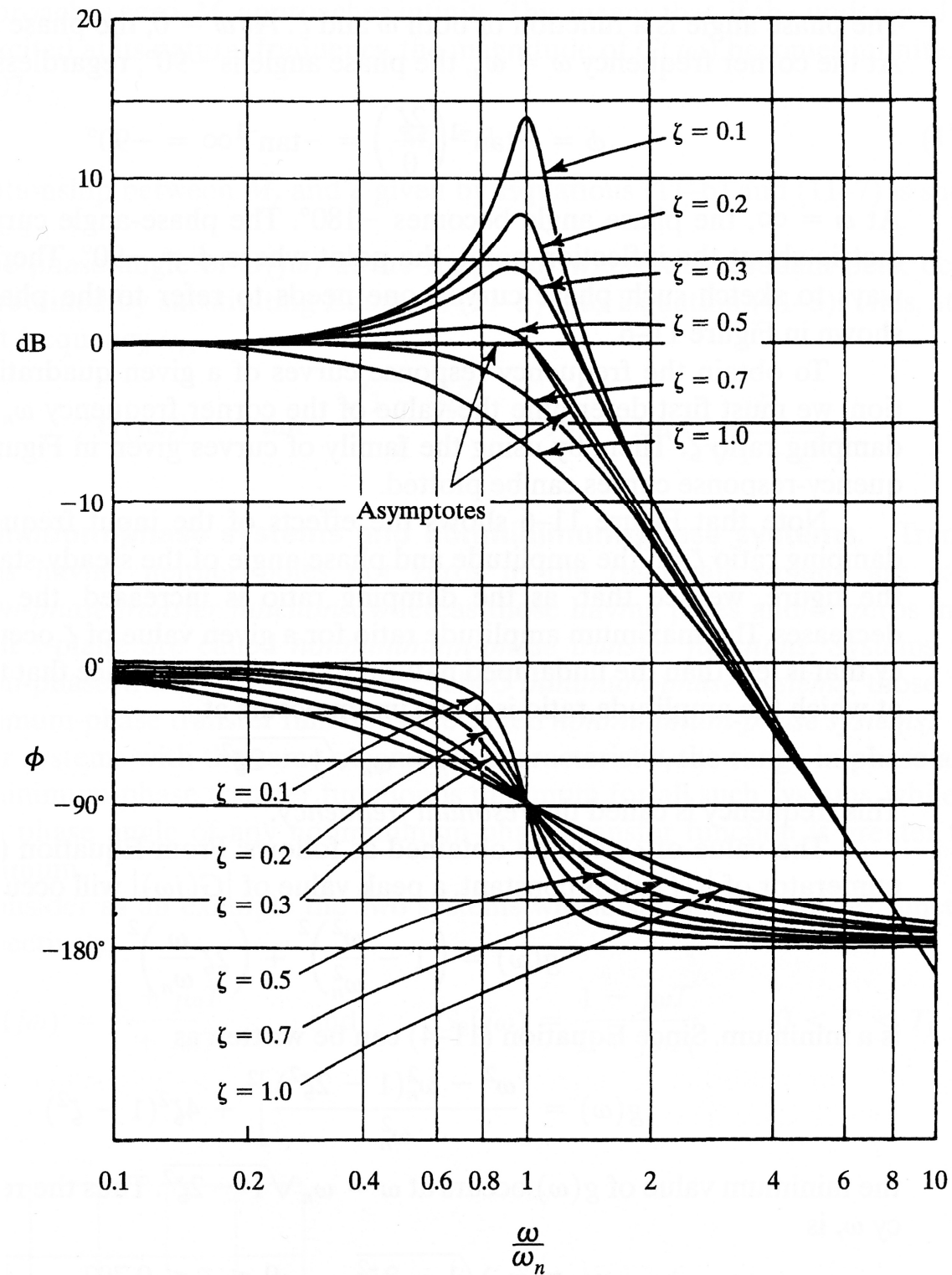
- **Steady-state** response of a (stable, linear, time-invariant) system to a sinusoidal input.
- Vary the frequency of the input signal over a certain range and study the resulting response.
- The output (response) will also be sinusoidal, but in general the amplitude and phase will be different from those of the input.



Obtain Frequency Response Experimentally



Bode Plot



Safety First!

Wear eye protection at all times!

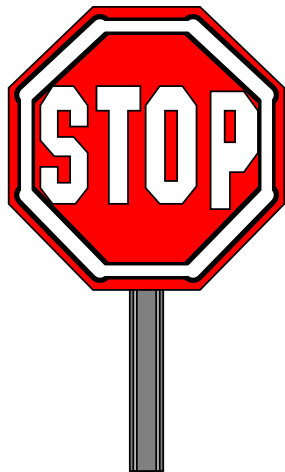
Follow GSI's instructions carefully!

Be aware of the experimental apparatus at all times...what is moving, what is hot, what is cold.

Do not exceed stated power and timing parameters!

Safety Precautions: glasses, shoes, hair, clothing!

Review and sign Safety sheet, return in first lab to GSI



& THINK!

